FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Main)-2023

TEST DATE: 27-03-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

2. C

3. A

Sol. $91.3\times V = 0.31$

4. A

Sol. $L^2\sqrt{2gy} = 2\pi R^2\sqrt{2gy} \Rightarrow R = \frac{L}{\sqrt{2n}}$

5. B

6. C

Sol. From 0 to t_1 emf is constant and $-ive$

From $t_1 to t_2$ emf is zero

From t_2 to $t_1 + t_2$ emf is constant and $+ive$

7. D

8. D

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Sol. $V=\frac{1}{4\pi \in_0}.\frac{\theta}{1+\frac{1}{2}}=\frac{\theta}{6\pi \in_0}$

9. A

10. C

Sol. $\frac{hc}{\lambda}-\phi=eV_{max}$

11. B

Sol. $\frac{t}{1/2}=\frac{3}{4}t\Rightarrow\frac{11}{2}t-4\Rightarrow\frac{3}{2}t=\frac{2t}{1/2}$

12. A

Sol. $rate\propto T^4$

13. B

Sol. $\frac{dr}{dt}=\text{constant}$

14. C

Sol. $\alpha=\frac{\omega dw}{dv}\Rightarrow at\,\theta=1\alpha=4\times3=12\,rod/\sec^2$

15. D

16. B

Sol. $2\pi\sqrt{\frac{L}{(mg-kv)/m}}=2\pi\sqrt{\frac{L}{g}}\times2$

$\Rightarrow\frac{1}{mg-kv}=\frac{4}{mg}\Rightarrow mg-kv=mg/4$

$v=\frac{3mg}{4k}$

17. D

18. C

19. C

20. B

Sol. Angle of incident u is greater than vertical angle

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SECTION – B

21. 2

Sol. $\dfrac{\sqrt{\dfrac{2GM_1}{R_1}}}{\sqrt{\dfrac{2GM_2}{R_2}}}=2$

$\dfrac{M_1R_2}{M_2R_1}=4=\dfrac{R_1^2}{R_2^2}\Rightarrow\dfrac{R_1}{R_2}=2$

22. 5

Sol. $\eta=\dfrac{\sum\Delta Q}{\Delta Q(+ve)}\times100=\dfrac{50R}{100R}\times100=50\%$

23. 1

Sol. mu-mv_1+nmv_2

$u=v_1+nv_2-----(1)$

$-1=\dfrac{v_1-v_2}{u}-----(2)$

Fractional transfer $=\dfrac{4n}{1+n^2}=y$

So $dy/dn=0\Rightarrow n=1$

24. 8

Sol. $X_{cm}=\dfrac{\int_0^{\ell}x\lambda_0(1+\alpha x)\,dx}{\int_0^{\ell}\lambda_0(1+\alpha x)\,dx}=\dfrac{5}{9}m$

$I_{cm}=\int_{-5/9}^{4/9}x^2\lambda_0(1+\alpha x)\,dx=\dfrac{211}{2^2\times3^6}kg-m^2$

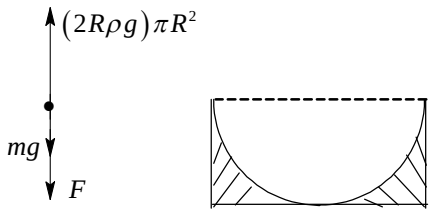
25. 5

Sol. Drawing F.B.D of water (shaded part)

$F=-mg+2\pi R^3\rho g$

$F=-\left(\pi R^3-\dfrac{2}{3}\pi R^3\right)\rho g+2\pi R^3\rho g$

$F=\dfrac{5}{3}\pi R^3\rho g$



26. 3

Sol. $(10\times11\text{-}10\times6)\times10^{-4}\times2T=\Delta w$

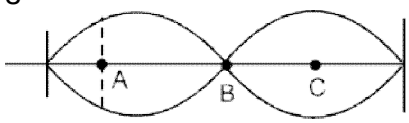
$50\times10^{-4}\times2T=3\times10^{-4}$

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$\Rightarrow T=\frac{3}{100}=3\times10^{-2}N/m$

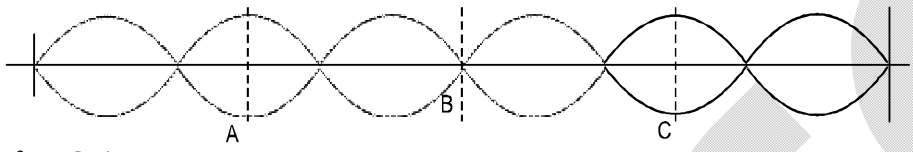
27. 3

Sol.



(n=2)

$f_1=\frac{2v}{2L}$ ----------(i)



$\frac{f_2}{f_1}=\frac{6v/2L}{\frac{2V}{2L}}=3$

$f_2=3f_1=3\times100=(300Hz)$

28. 0

29. 9

30. 11

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Chemistry

PART – B

SECTION – A

31. A
Sol. Down the group covalent character decrease

32. C

33. A

34. D

35. B

36. C

37. B
Sol. Using MOT (paramagnetic entities are attracted towards externally applied magnetic field) find paramagnetic ions.

38. B

39. C
Sol. Rate of effusion $\propto \frac{1}{\sqrt{T}}$

40. D

41. B
Sol. Charge is not balanced in reaction except option (B)

42. A

43. B

44. B

45. A

46. D
Sol. $Zn + 2NaOH \rightarrow \underset{(A)}{Na_2ZnO_2} + \underset{(X)}{H_2}$

$Na_2ZnO_2 + H_2S \rightarrow Zns + 2NaOH$

47. C

48. C
Sol. Zn^{2+}, Li^{+}, K^{+}, Al^{3+} are colourless where as Co^{2+} is coloured.

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49. D

50. B

SECTION – B

51. 2

Sol. $Na\left[\overset{+I}{A}g(CN)_2\right]$

52. 5

53. 5

Sol. Area under the curve $= -1\times3\times10^{-5} - \frac{1}{2}\times3\times3\times10^{-5} = -7.5\times10^{-5}\ J$

54. 25

Sol. $P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 20\,NO_2\uparrow + 4H_2O$

$m.moles\ of\ NaOH = 2\times m.moles\ of\ H_3PO_4 \Rightarrow m.moles\ H_3PO_4 = \frac{100\times0.1}{2} = 5$

$\text{m.moles of } HNO_3 = \frac{20}{4}\times5 = 25$

Volume of $HNO_3 = 25\,\text{ml}$

55. 5

56. 3

57. 22

58. 1

59. 2

60. 700

Sol. $X_x = 0.30\ \&\ Y_x = 0.60$

$P_B^0 = 700\,\text{mm}$

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Mathematics

PART – C

SECTION – A

61. D

Sol. $f(x)=x^4+2x^3-x^2+1$ is polynomial of even degree hence its range can't be *R*. Hence not surjective.

$f(x)=x^3+x+1,\ f'(x)=3x^2+1 \Rightarrow$ monotonic hence bijective.

$f(x)=\sqrt{1+x^2}$, neither surjective not injective.

$f(x)=x^3+2x^2-x+1,\ f'(x)=3x^2+4x-1 \Rightarrow$ $D = 16 + 12 > 0$.

Hence $f(x)$ is non-monotonic cubic polynomial hence surjective but not injective.

62. B

Sol. For $0 < x < \frac{\pi}{2}$ or $\frac{\pi}{2} < x < \pi$, $0 < \sin x < 1$.

$\therefore$ for $n > 1$, $\sin x > \sin^n x$

$\therefore$ $f(x)=\left[\frac{3(\sin x-\sin^n x)}{\sin x-\sin^n x}\right]=3$, $x \neq \frac{\pi}{2}$ and $f(x) = 3$, $x=\frac{\pi}{2}$

Thus in $(0, \pi)$, $f(x) = 3$.

Hence $f(x)$ is continuous and differentiable at $x=\frac{\pi}{2}$.

63. C

Sol. If $x_0 = K \in I$, then $\lim\limits_{x\to K+} f(x)=\lim\limits_{x\to K} K+\sqrt{x-K}=K$

and $\lim\limits_{x\to K-} f(x)=\lim\limits_{x\to K-}(K-1)+\sqrt{x-(K-1)}$

$= \lim\limits_{x\to K}(K-1)+\sqrt{x-K+1}=K-1+1=K$

Hence f is continuous at every $x_0 = K \in I$. If $x_0 \in R \sim I$, then $[x]$ is continuous at x_0 which is turn gives that f is continuous at every $x_0 \in R \sim I$

64. B

Sol. $f(x)=\int\frac{x^2(\sqrt{1+x^2}-1)}{(1+x^2)(1+x^2-1)}dx=\int\frac{\sqrt{1+x^2}-1}{1+x^2}dx=\int\frac{dx}{\sqrt{1+x^2}}-\int\frac{dx}{1+x^2}$

$$=\log(x+\sqrt{1+x^2})-\tan^{-1}x+k$$

$\therefore$ $f(0)=\log 1-\tan^{-1}0+k=k=0$

$\therefore$ $f(x)=\log(x+\sqrt{1+x^2})-\tan^{-1}x$

$\therefore$ $f(1)=\log(1+\sqrt{2})-\frac{\pi}{4}$

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65. A

Sol. $p \to q$ must be true & q→r must be false

$\therefore$ q must be true, r must be false but p can be true or false.

66. B

Sol. Let $I = f\left(\frac{1}{2}\right) - f\left(\frac{1}{3}\right) = \int_0^{\pi/4} \log_e \left(\frac{1+\frac{1}{2}\tan z}{1+\frac{1}{3}\tan z} \right) dz = \int_0^{\pi/4} \log_e \left(\frac{3}{2} . \frac{2+\tan z}{3+\tan z} \right) dz$

$$= \int_0^{\pi/8} \left(\log_e \left(\frac{3}{2} . \frac{2+\tan z}{3+\tan z} \right) + \log_e \left(\frac{3}{2} . \frac{2+\tan\left(\frac{\pi}{4}-z\right)}{3+\tan\left(\frac{\pi}{4}-z\right)} \right) \right) dz = \frac{\pi}{8} \log_e \left(\frac{9}{8} \right)$$

$$\Rightarrow \quad I = \frac{\pi}{8} \log_e \left(\frac{9}{8} \right)$$

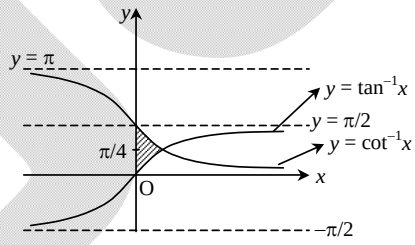
67. D

Sol. Required area,

$$\Delta = \int_0^{\pi/4} \tan y \, dy + \int_{\pi/4}^{\pi/2} \cot y \, dy$$

$$= \ln \sec y \Big|_0^{\pi/4} + \ln \sin y \Big|_{5\pi/4}^{\pi/2}$$

$$= \ln \sqrt{2} - \ln \frac{1}{\sqrt{2}} = \ln 2 .$$



68. A

Sol. Put $\tan^{-1} x = \theta$ $\tan^{-1} x = \theta \Rightarrow \theta \in (-\pi/2, -\pi/4) \Rightarrow 2\theta \in (-\pi, -\pi/2)$

$\therefore$ equation becomes $\pi - \cos^{-1}(\cos 2\theta) + \sin^{-1}(\sin 2\theta) + \tan^{-1}(\tan 2\theta) = \pi/3$

On solving we get $\theta = -\pi/3 \Rightarrow x = -\sqrt{3}$

69. B

Sol. Numerator = $\cos 6x + \cos 4x + 5(\cos 4x + \cos 2x) + 10(\cos 2x + 1)$

= $2\cos x\{\cos 5x + 5\cos 3x + 10\cos x\}$

$\therefore y = \frac{\text{Nr}}{\text{Dr}} = 2\cos x, \ \frac{dy}{dx} = -2\sin x$

70. C

Sol. Adding the given two

$\sin(B+C) = 1 \ \Rightarrow \ B + C = 90°$

Subtracting, $\sin(B-C) = \frac{\sqrt{3}-2}{\sqrt{3}} \neq 0$

$\therefore B \neq C$

$\therefore$ Triangle is right-angled.

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71. B

Sol. Let A_1 be the property that HIN appears

Let A_2 be the property that DUS appears

Let A_3 be the property that TAN appears

$n(A_1) = 7! = 5040$

$n(A_2) = \frac{7!}{2!} = 2520$

$n(A_3) = 7! = 5040$

$n(A_1 \cap A_2) = n(A_2 \cap A_3) = n(A_3 \cap A_1) = 5! = 120$

$n(A_1 \cap A_2 \cap A_3) = 3! = 6$

$n(A_1 \cup A_2 \cup A_3) = \sum n(A_i) - \sum_{i \neq j} n(A_i \cap A_j) + n(A_1 \cap A_2 \cap A_3) = 12246$

Total number of ways to arrange the letters of the word HINDUSTAN $= \frac{9!}{2!} = 181440$

$\therefore$ Number of ways in which neither the pattern HIN or DUS and TAN appears
= 181440 – 12246 = 169194

72. A

Sol. Total number of matrices = 3^9
Number of symmetric matrices = 3^6
Number of skew symmetric matrices = 3^3
The zero matrix is both symmetric and skew-symmetric the required probability is

$= \frac{3^9 - 3^6 - 3^3 + 1}{3^9}$

73. D

Sol. $(1 - x + x^2)^n = a_0 - a_1 x + a_2 x^2 - .... + a_{2n} x^{2n}$

$(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + ... + {}^nC_n x^n$

On multiplying and collecting coefficient of x^{3r}, we get $\sum_{k=0}^{3r} (-1)^k a_k \, {}^nC_{3r-k} = {}^nC_r$

74. A

Sol. We can write Δ as

$$\Delta = \begin{vmatrix} 0 & \sin x - \cos x & \log \tan x \\ -(\sin x - \cos x) & 0 & -\log \tan x \\ -\log \tan x & \log \tan x & 0 \end{vmatrix}$$

Which is determinant of a skew symmetric matrix of odd order, hence value will be zero.

75. C

Sol. By the given condition $(1 - ab)m^2 - (a^2 + b^2)m - (1 + ab) = 0$

$\Rightarrow m(a^2 + b^2) + (m^2 + 1)ab = m^2 - 1 \quad \ldots$(i)

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Now H_1 = First H.M. between *a* and *b* = $\frac{(m+1)ab}{a+mb}$ and $H_m = \frac{(m+1)ab}{b+ma}$

$\therefore\ H_m - H_1 = (m+1)ab\left[\frac{1}{b+ma} - \frac{1}{a+mb}\right] = (m+1)ab\frac{[(m-1)(b-a)]}{(b+ma)(a+mb)}$

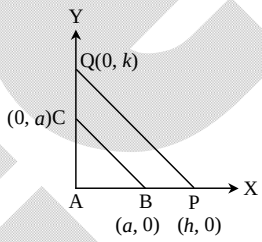
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$\frac{(m^2-1)ab(b-a)}{m(a^2+b^2)+(m^2+1)ab} = \frac{(m^2-1)ab(b-a)}{m^2-1}$ [From (i)]

= $ab(b-a)$.

76. C

Sol. We take *A* as the origin and *AB* and *AC* as *x*-axis and *y*-axis respectively. Let *AP* = *h*, *AQ* = *k* Equation of the line *PQ* is



$\frac{x}{h} + \frac{y}{k} = 1$ …(i)

Given, *BP* . *CQ* = AB^2

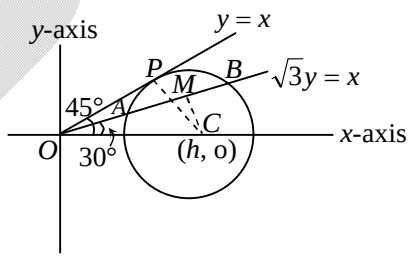
$\Rightarrow (h-a)(k-a) = a^2 \Rightarrow hk - ak - ah + a^2 = a^2$ or $ak + ha = hk$

Or $\frac{a}{h} + \frac{a}{k} = 1$ ….(ii)

From (ii), it follows that line (i) i.e. PQ always passes through a fixed point (*a*, *a*).

77. A

Sol. $OP = CP = \frac{h}{\sqrt{2}} = AC$



$CM = h\sin 30° = \frac{h}{2}$

From Δ *ACM*

$AC^2 = AM^2 + CM^2$

$\frac{h^2}{2} = 1 + \frac{h^2}{4} \Rightarrow \frac{h^2}{4} = 1 \Rightarrow h = 2$

radius $= \sqrt{2}$

$(x-2)^2 + y^2 = 2; x^2 + y^2 - 4x + 2 = 0$

78. B

Sol. Mid-point of (1, 2) and (3, 6) is (2, 4), line passing through (1, 6) and (2, 4) is parallel to axis of the parabola (property). So, slope of its axis is –2 or slope of its directrix is 1/2, also in this case tangents are perpendicular. So, (1, 6) must lie on directrix,

$\therefore\ y - 6 = \frac{1}{2}(x-1) \Rightarrow x - 2y + 11 = 0$

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79. C

Sol. $b^2 = d_1 d_2$, where d_1 and d_2 are perpendicular distances of any tangent from focii of the hyperbola $b^2 = 4 \times 2 = 8 \Rightarrow b = 2\sqrt{2}$

$2ae = \sqrt{72} = 6\sqrt{2}$

So $a^2 = 10$

Centre is (5, 1)

Equation of conjugate axis $x - y = 4$ and equation of transverse *axis* is $x + y = 6$

$\therefore$ Equation of hyperbola is $\dfrac{(x-y-4)^2}{20} - \dfrac{(x+y-6)^2}{16} = 1$

80. B

Sol. $\dfrac{\sin \pi/12}{\sin x} + \dfrac{\cos \pi/12}{\cos x} = 2 \Rightarrow \sin\dfrac{\pi}{12}\cos x + \cos\dfrac{\pi}{12}\sin x = 2\sin x \cos x$

$\sin\left(\dfrac{\pi}{12} + x\right) = \sin 2x$

$\dfrac{\pi}{12} + x = 2x$ and $\dfrac{\pi}{12} + x = \pi - 2x$

$\Rightarrow x = \dfrac{\pi}{12}$ and $x = \dfrac{11\pi}{36}$

SECTION – B

81. 1

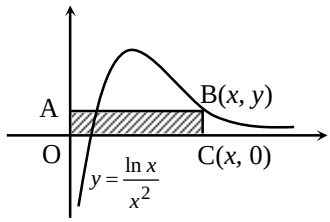
Sol. Area = $A = xy$

$A = \dfrac{\ln x}{x} \Rightarrow \dfrac{dA}{dx} = \dfrac{1 - \ln x}{x^2}$

$\dfrac{dA}{dx} = 0 \Rightarrow x = e$

$\dfrac{dA}{dx} < 0, \forall x > e$ and $\dfrac{dA}{dx} > 0, \forall x < e$.

Hence $x = e$ is point of maxima. $A_{max} = e.\dfrac{1}{e^2} = \dfrac{1}{e}$



82. 2

Sol. Put $\tan y = z \quad \therefore \quad \sec^2 y \dfrac{dy}{dx} = \dfrac{dz}{dx} \quad \therefore \quad \dfrac{dz}{dx} + xz = x^3$

which is a linear differential equation, on solving we get

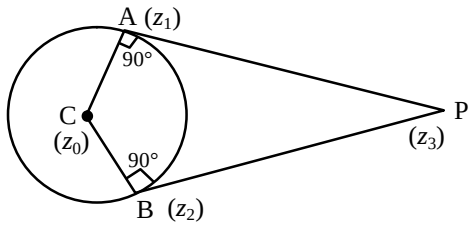
$\tan y = x^2 - 2 + ce^{-x^2/2}$

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83. 1

Sol. Let z_1, z_2 are represented by *A*, *B* whereas z_0 is represented by *C*.

Let *P* represent z_3



$$\frac{z_3 - z_1}{z_0 - z_1} = \frac{PA}{AC} e^{\frac{i\pi}{2}} ; \frac{z_0 - z_2}{z_3 - z_2} = \frac{BC}{PB} e^{\frac{i\pi}{2}}$$

$$\frac{z_3 - z_1}{z_0 - z_1} . \frac{z_0 - z_2}{z_3 - z_2} = \frac{PA}{\text{radius}} . \frac{\text{radius}}{PB} e^{i\pi} = -1 \quad (\because PA = PB)$$

84. 1

Sol. LHS achieves its maximum value at $x = 1$ (in the domain of the equation i.e., $x > 0$)
Maximum value of LHS =1 at $x = 1$
RHS achieves it minimum value at $x = 1$ (in the domain of equation i.e., $x > 0$)
Minimum value of RHS = 1 at $x = 1$
Hence only possible value of x satisfying the given equation is 1.

85. 1

Sol. $A\,(\text{Adj } A) = |A| I_3$

$$K = |A| = 2(\sin^6 x + \cos^6 x) - 3(\sin^4 x + \cos^4 x)$$

$$= 2(1 - 3\sin^2 x . \cos^2 x) - 3(1 - 2\sin^2 x \cos^2 x) = -1$$

86. 2

Sol. We have, $a_n = S_n - S_{n-1} \ldots$(i)

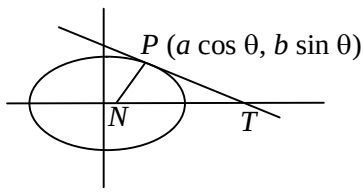
and $a_{n-1} = S_{n-1} - S_{n-2} \ldots$(ii)

$\therefore d = a_n - a_{n-1} = (S_n - S_{n-1}) - (S_{n-1} - S_{n-2})$ [From (i) and (ii)]

$= S_n - 2S_{n-1} + S_{n-2}$

87. 6

Sol. $e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{3}{5}$



$$T \equiv \left(\frac{a}{\cos\theta}, 0\right), \; N \equiv \left(\frac{a^2 - b^2}{a\sec\theta}, 0\right)$$

On solving we get, $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$

$\therefore$ Area of $\Delta PSS' = 6\sqrt{3}$ sq. units

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88. 1

Sol. The expression = $\sin^2 A + 2\sin A\cos B\sin C + \cos^2 B$

$= \sin^2 A + 1 - \sin^2 B + 2\sin A\cos B\sin C$

$= \sin(A+B)\sin(A-B) + 1 + 2\sin A\cos B\sin C$

$= -\sin C\sin(A-B) + 2\sin A\cos B\sin C + 1$

$= \sin C[2\sin A\cos B - \sin(A-B)] + 1$

$= \sin C\sin(A+B) + 1$

$= \sin C(-\sin C) + 1 = \cos^2 C$

89. 32

Sol. The equation of a plane parallel to the plane $\vec{r}\,.(4\hat{i} - 12\hat{j} - 3\hat{k}) - 7 = 0$ is $\vec{r}\,.(4\hat{i} - 12\hat{j} - 3\hat{k}) + \lambda = 0$. This passes through $2\hat{i} - \hat{j} - 4\hat{k}$. Therefore

$(2\hat{i} - \hat{j} - 4\hat{k})\,.(4\hat{i} - 12\hat{j} - 3\hat{k}) + \lambda = 0$

$\Rightarrow 8 + 12 + 12 + \lambda = 0 \quad \Rightarrow \quad \lambda = -32$

So, the required plane is $\vec{r}\,.(4\hat{i} - 12\hat{j} - 3\hat{k}) - 32 = 0$

90. 7

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